



Hacking Bloom's Taxonomy:

A Pyramid Scheme

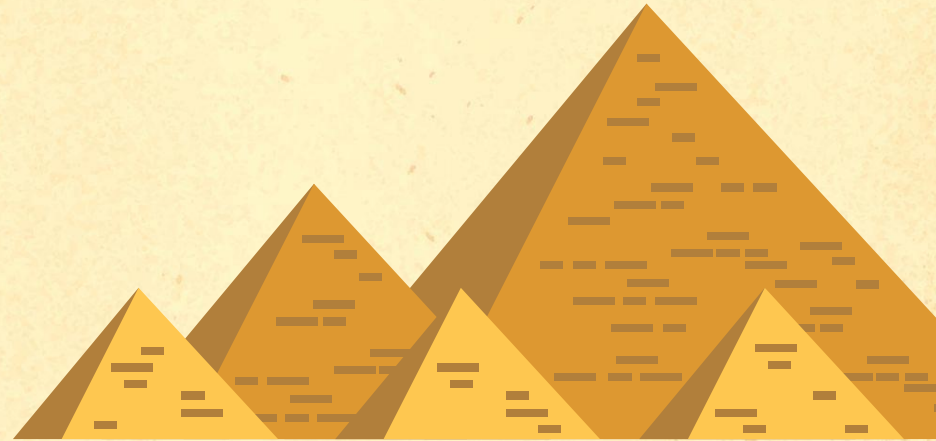




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01

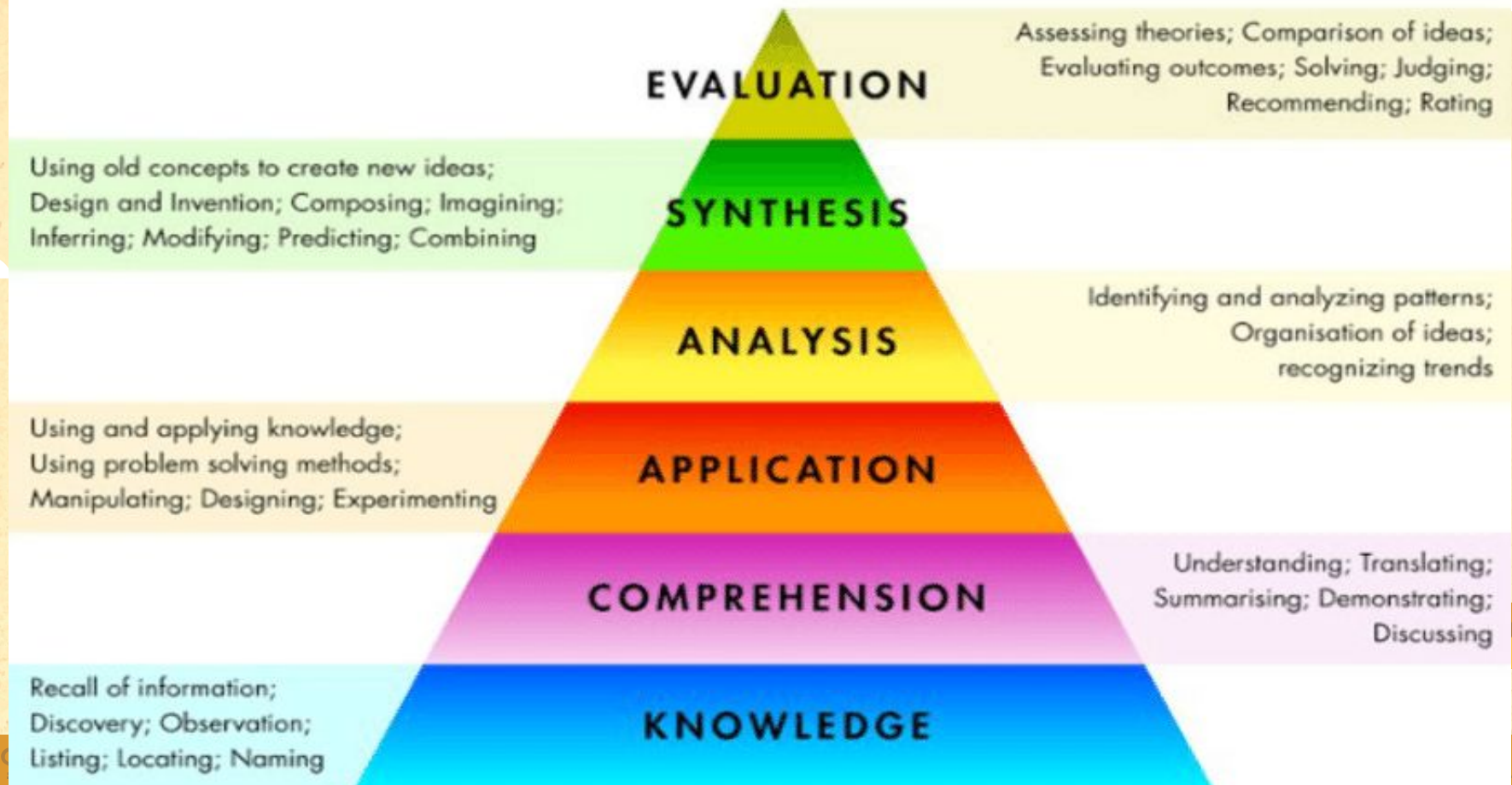
Introduction

1. What is Bloom's Taxonomy (BT)?
2. What is teachers' background knowledge of BT?
3. How can we build on teachers' knowledge of BT to increase student learning and motivation?

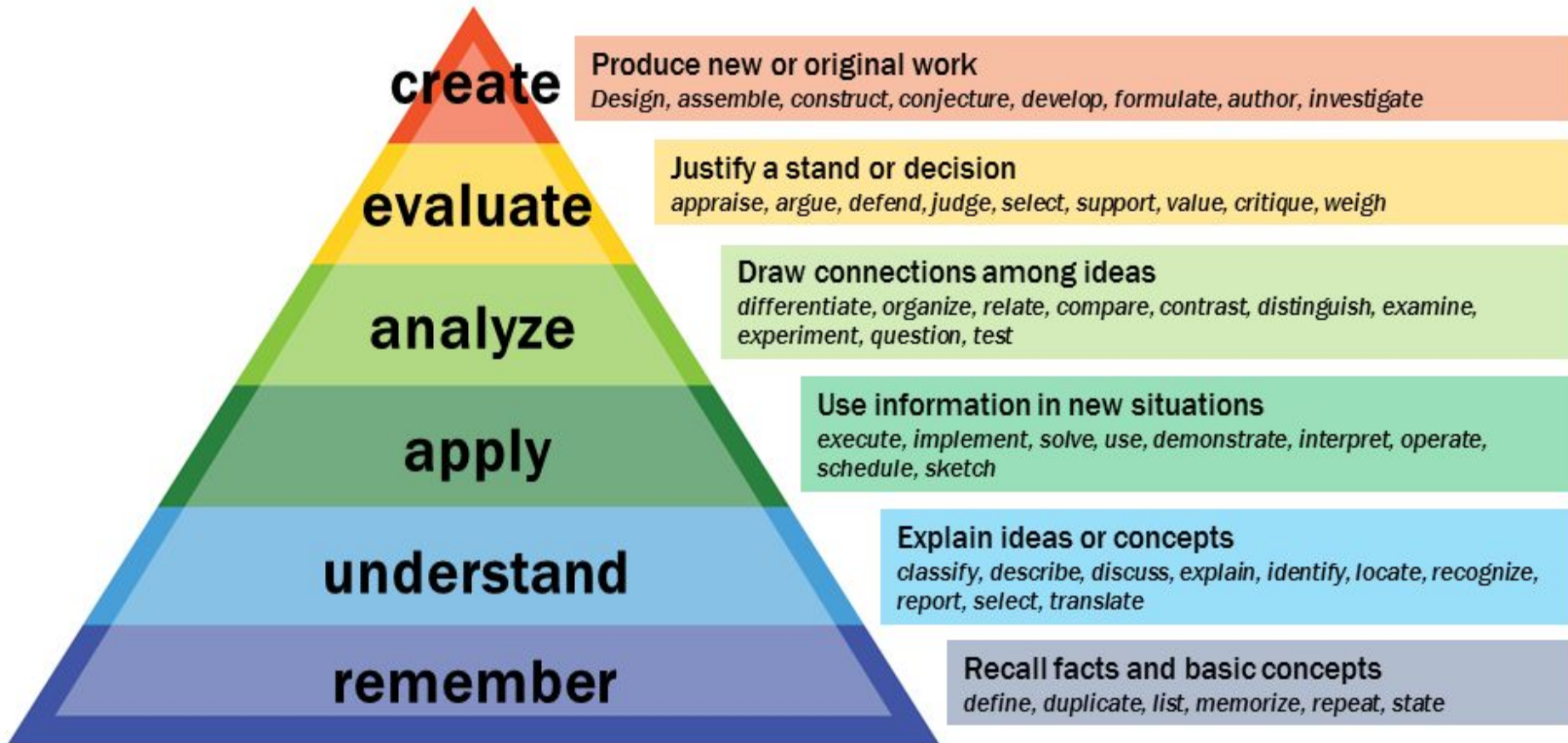


BLOOMS TAXONOMY

Citation: DeFalco, Jeanine. (2018). Standardizing Metadata Tagging with Pedagogical Identifiers.



Bloom's Taxonomy



THE **NEW*** BLOOM'S TAXONOMY & FOREIGN LANGUAGE INSTRUCTION

GOOD COMPREHENSIBLE INPUT-BASED SECOND LANGUAGE TEACHING USES ALL OF THESE LEVELS OF THINKING
We can encourage students to think at high levels even in beginning courses.

Level of Thinking	Key Question / Verb Examples	Examples of Comprehensible Input-Based Activities
1. REMEMBER <i>(lowest level)</i> <i>Retrieving, recognizing, and recalling relevant knowledge from long-term memory.</i>	Can the student recall or remember the information? arrange, define, label, list, match, memorize, name, order, recall, recognize, repeat, reproduce, restate, state	*Answer questions that begin with who, what, when, where (if the answer is explicitly given in a story) *Match characters to action/dialogue *Information gap questions *True-False, Either/Or statements *Match L2 vocabulary to English
2. UNDERSTAND <i>Constructing meaning from oral, written, and graphic messages.</i>	Can the student explain ideas or concepts? classify, compare, describe, discuss, explain, express, give examples, give main idea, infer, interpret, paraphrase, report, review, select, summarize, translate	*Summarize a story in own words *Restate main idea of story *Explain why a character in a story does/says something (when answer was stated in story) *Describe a person/place in the story *Translate text aloud to English
3. APPLY <i>Carrying out or using a procedure.</i>	Can the student use the information in a new way? apply, choose, demonstrate, dramatize, execute, illustrate, implement, interpret, outline, point out, role play, show, sketch, solve, use	*Act out novel commands *Rewrite a story from a different point of view (POV) *Act out a story *Draw a story
4. ANALYZE <i>Breaking material into constituent parts, determining how the parts relate to one another and to an overall structure or purpose.</i>	Can the student distinguish between the different parts? analyze, appraise, attribute, break down, calculate, categorize, compare, contrast, differentiate, , discriminate, dissect, distinguish, examine, organize, question, test	*Answer <u>why</u> or <u>open-ended</u> questions (when answer is indirectly stated or implied in a story) *Break down the main actions of the story *Use a VENN diagram to compare and contrast (characters, situations, countries, cultures, schools, etc.)
5. EVALUATE <i>Making judgments based on criteria and standards.</i>	Can the student justify a stand or decision? argue, appraise, assess critique, check, conclude, compare, criticize, defend, estimate, evaluate, judge, justify, predict, rate, select, support, value	*Evaluate appropriate/inappropriate actions of characters *Compare cultures *Predict what will happen next *Make inferences
6. DESIGN <i>(highest level)</i> <i>Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern.</i>	Can the student create a new product or point of view? assemble, combine, compile, compose, create, construct, design, develop, devise, formulate, generate, invent, organize, plan, prepare, produce, propose, reconstruct, revise, rewrite, write	*Create and give novel commands *Write an original story *Compose a class story *Invent new details for a story *Generate / invent answers to hypothetical questions *Rewrite a story adding details &/or characters that were not in the original

* Alan Bloom's classic 1956 learning taxonomy was revised and refined by Lorin Anderson and David Krathwohl in 2000.



02

Rationale

How does Bloom's Taxonomy increase student motivation? Why should we care?



Some Empirical Evidence



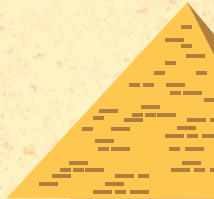
Observations & Anecdotes

- *Student complaint #1:* Tests only assess memorization
- *Student complaint #2:* Teachers give too much pointless busywork
- *Teacher complaint:* Many students, though gifted, seem to lack motivation



Academic Literature

- Framework for a type of differentiated instruction (Rule & Lord, 2003)
- Can be integrated with Gardner's multiple intelligences (Noble, 2004)
- Addresses depth, complexity, challenge, and abstraction (Little, 2012)



More Empirical Evidence

Bloom et al. maintain ...

“[synthesis activities] can become highly absorbing, more so than the usual run of school assignments” (Bloom et al., 1956, pp. 167-168).



Dylan Hyman gives a TEDx talk on higher order thinking in the classroom (TEDx Talks, 2015).



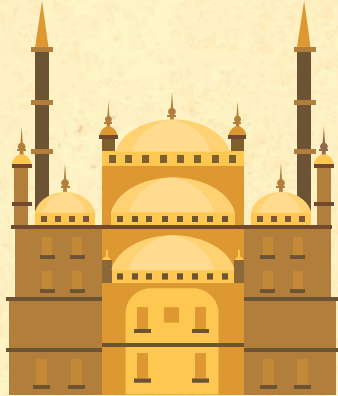
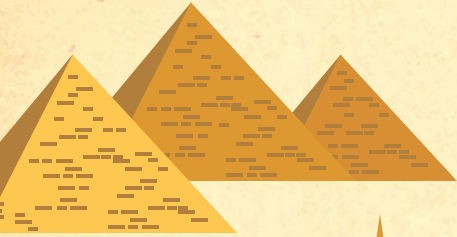
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Synchronizing Our Efforts

How do we hack Bloom's Taxonomy as a professional learning community across disciplines?



Planning & Facilitating the Workshop



Form Cohorts

- Engage in cross-curricular lesson planning
- Use document sharing for flexible collaboration



Icebreaker Ideas

- 'Group Story' or 'Common Chain'
- BT sorting activity in small groups



Close with Review

- PL version of the Pyramid game show





04

In the Classroom

What does transdisciplinary implementation look like across disciplines?



Interdisciplinary & Collaborative

- Within the PLCs, educators can partner with a colleague who teaches in a different discipline and brainstorm ways to make their lessons interdisciplinary.
- In the same pairs, educators can develop a lesson plan that includes learning objectives using keywords from B.T. starting at the bottom level and ending at the top



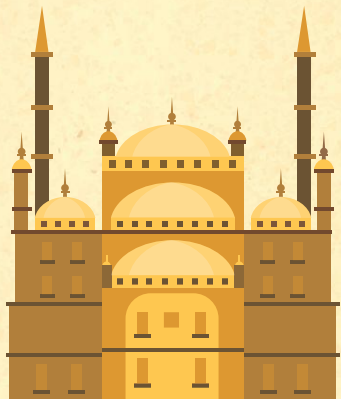
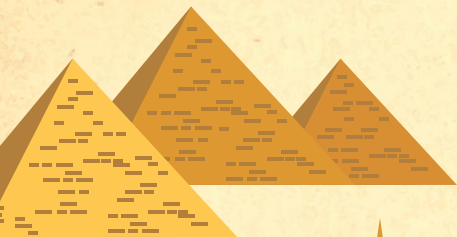
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Resources

What will it take to pull off this pyramid scheme?



What We Need



Space & Time

- Classroom or high school ilab,
- 1x per month for 4 months
- 1 hour per workshop



Human Resources

- IT Team
- Project Manager



Funds

- ~ \$600 to cover light refreshments for 4 meetings





06

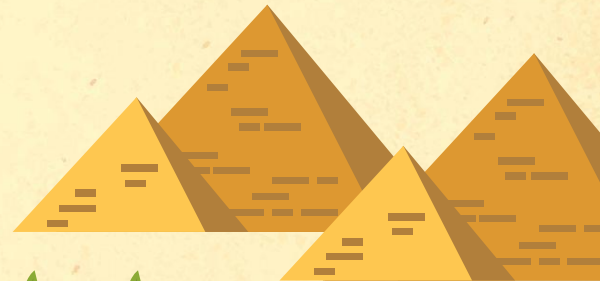
Closing

Final thoughts on hacking Bloom's
Taxonomy





Thanks!



Questions or comments?

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Let's connect!

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